hydrocarbon tail linked to one end of the steroid and a hydroxyl group at C3 and a double bond between C5 and C6. The hydroxyl group at C3 constitutes the polar head. The rest of the molecule is hydrophobic.

*Normal value of total blood cholesterol level is < 200 mg/dL.

Properties of cholesterol:

i. Cholesterol exists as crystals that are white, shiny and rhombic in nature

ii. It is tasteless and odorless

iii. It has a high melting point of 150°C

iv. It is insoluble in water and soluble in fat solvents

v. It is a poor conductor of heat and electricity and serves as an insulator. In brain, where it is present abundantly, it acts as an insulator against nerve impulse which is electric in nature

vi. Cholesterol, when oxidized under suitable conditions, undergoes rapid oxidation to form a ketone called cholestenone

vii. The hydroxyl group of cholesterol readily forms ester with fatty acids like stearic acid

Cholesterol is essential to life, as it performs several important functions

i. lt is a structural component of the cell membrane

ii. Cholesterol is the precursor for the synthesis of all other steroids in the body. These include steroid hormones, vitamin D and bile acids

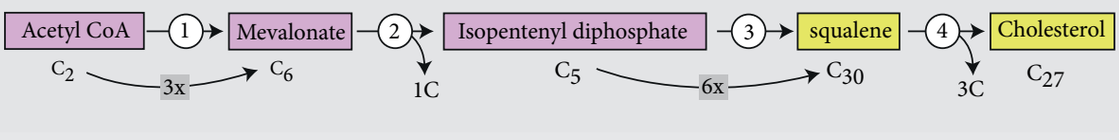
iii. lt is an essential ingredient in the structure of lipoproteins which transport lipids in the body. Fatty acids are transported to liver as cholesteryl esters for oxidation.

5.4.1 Biosynthesis of cholesterol

About 1 g of cholesterol is synthesized per day in adults. Almost all the tissues of the body participate in cholesterol biosynthesis. The largest contribution is made by liver (50%), intestine (15%), skin, adrenal cortex and reproductive tissue.

The enzymes involved in cholesterol synthesis are found in the cytosol and microsomal fractions of the cell. Acetate of acetyl CoA provides all the carbon atoms in cholesterol. The reducing equivalents are supplied by NADPH while ATP provides energy.

For the production of one mole of cholesterol, 18 moles of acetyl CoA, 36 moles of ATP and 16 moles of NADPH are required.



The four basic steps in cholesterol synthesis are,

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